

Teacher Guide: Week 1

Logic Is Not Magic: What Logic Can and Cannot Do

10th Grade CUE Test Run Based on Richard Whately's *Elements of Logic*

Week 1 Overview

This first week introduces Richard Whately's basic account of logic. Students should leave the week understanding that logic is not a machine that automatically discovers truth. Logic studies the structure of reasoning and helps expose bad inferences. This establishes the foundation for later weeks on terms, propositions, arguments, fallacies, ambiguity, induction, and verbal versus real disputes.

Essential Question

What can logic actually do for a thinker, and what should we not expect it to do?

Student-Facing Materials

- *Logic Is Not Magic: What Logic Can and Cannot Do* — student reading handout.
- *Week 1 Argument Lab: What Follows From What?* — practice handout and window-note task.

Learning Goals

Students will be able to:

1. Define logic as the study of reasoning and argument structure.
2. Distinguish what logic can do from what it cannot do.
3. Identify conclusion, stated reason, and hidden assumption in a simple argument.
4. Explain why confidence, popularity, age, or emotion do not by themselves prove a claim.
5. Write a short claim paragraph using an analogy for logic.

Key Vocabulary

logic, reasoning, argument, premise, conclusion, deduction, fallacy, science, art, panacea, hidden assumption

Weekly Lesson Sequence

Day 1: Opening Question and First Read

- Begin with the question: *If someone is good at arguing, does that mean he is good at finding truth?*
- Have students answer briefly in writing.
- Introduce Whately as a nineteenth-century thinker trying to rescue logic from both overpraise and contempt.
- Read the introduction and first half of the student reading aloud or in pairs.
- Stop after the analogy paragraph. Ask: What does Whately think logic is being unfairly expected to do?

Day 2: Finish Reading and Annotate Whately's Claim

- Finish the reading.
- Students mark three places: one sentence showing what logic can do, one showing what it cannot do, and one example that separates conclusion, stated reason, hidden assumption, and test.
- Complete Reading Focus questions 1–6.

Day 3: Argument Skeleton Practice

- Introduce the pattern: conclusion, stated reason, hidden assumption, test.
- Work through the modeled example from the Argument Lab handout.
- Students complete the fresh Part II arguments individually, then compare answers with a partner.
- Emphasize that Whately's basic question is: *What follows from what?*

Day 4: Repair and Create Arguments

- Students choose one weak argument from the lab and repair it by narrowing the claim or strengthening the support.
- Students create an original test-case argument and identify its conclusion, stated reason, hidden assumption, and logical strength.
- Discuss several student examples. Push students to separate proof from confidence, tradition, honor, popularity, and impressive language.

Day 5: Exit Claim Paragraph

- Students write the exit claim paragraph from the revised lab.
- Require one reference to Whately's view and one example from school, literature, public speech, or everyday life.
- Collect the student reading and argument lab as the week's work product.

Suggested Board Notes

- Logic does not discover every truth.
- Logic studies whether a conclusion follows from premises.
- Bad facts can still produce bad conclusions even when the reasoning looks neat.
- Strong feelings, popularity, age, confidence, and repetition are not the same as proof.
- The first logical question: What follows from what?

Reading Focus Answer Guide

1. Main claim: Logic is the science and art of reasoning; it studies argument structure and helps guard against bad deductions.
2. Can do: examine whether conclusions follow; expose skipped steps, changed terms, hidden assumptions, or bad inferences.
3. Cannot do: supply facts by itself; make foolish people wise; discover every truth; replace observation, experience, memory, imagination, or moral judgment.
4. Four steps: find the conclusion, find the stated reason, identify the hidden assumption, and test whether the conclusion really follows.
5. Answers will vary. Strong responses should use one reading example and correctly name its conclusion, reason, assumption, and test. For example, the Macbeth argument concludes that Macbeth is not responsible; its stated reason is the witches' prediction; its hidden assumption is that prediction removes responsibility; the conclusion does not necessarily follow.
6. People blamed logic because some defenders promised too much for it, so critics rejected it when it failed to perform impossible tasks.

Argument Lab Answer Guide

The revised lab keeps one modeled argument from the reading, then moves students into fresh test cases. Strong student answers should not merely label parts; they should explain whether the reason actually earns the conclusion.

Argument 1: Statue and Noble Life

- Conclusion: The man must have lived nobly.
- Stated reason: The city placed a bronze statue of him in the town square.
- Hidden assumption: Public honor always goes to people who lived nobly, or a statue proves virtue.
- Test: The conclusion does not necessarily follow. A statue proves that a community chose to honor someone; it does not by itself prove that the person deserved the honor.

Argument 2: Difficult Words and Better Reasoning

- Conclusion: The first essay must have better reasoning.
- Stated reason: The first essay uses more difficult words than the second essay.
- Hidden assumption: More difficult language means better reasoning.
- Test: The conclusion does not follow. Difficult words may hide weak reasoning, and plain words may express strong reasoning.

Argument 3: Always Done This Way

- Conclusion: The old method must be the best method.
- Stated reason: The class has always done the assignment this way.
- Hidden assumption: Whatever has been done for a long time is best.
- Test: The conclusion does not necessarily follow. Tradition may preserve wisdom, but length of use alone does not prove superiority.

Argument 4: Same Action, Different Punishment

- Conclusion: The rule is not just.
- Stated reason: A just rule should treat the same action the same way, but this rule punishes one student and excuses another for the same action.
- Hidden assumption: Relevant circumstances are truly the same; no morally important difference justifies the different treatment.
- Test: This is a stronger argument if the cases really are alike in all relevant ways. If important circumstances differ, the argument needs more information.

Repair and Original Test Case

- Repaired arguments should either narrow an overclaim or add a reason that directly supports the conclusion.
- Original student arguments should include a clear conclusion and a stated reason. Stronger responses will identify a real hidden assumption rather than repeating the stated reason.

Assessment

Use the repaired argument, original test case, and exit paragraph as formative assessment. Strong work should:

- identify conclusions and reasons accurately;
- state hidden assumptions in complete, testable form;
- explain whether the reason actually earns the conclusion;
- improve a weak argument by narrowing the claim or strengthening the support;
- avoid treating impressive language, tradition, honor, or confidence as automatic proof.

Next Week Preview

Week 2 should move from reasoning in general to **terms**: the words that carry arguments. The next guiding question should be: *Why do arguments become unstable when key words are unclear?*